Ahsan Habib

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Education

BRAC University Dhaka, Bangladesh
B.Sc. in Computer Science and Engineering *Expected graduation: Fall 2026*

- CGPA: **3.76/4.00**; **3.95** in 300–400 level CSE courses
- **Vice Chancellor's List:** Summer 2024, Fall 2025 and Spring 2026 (semester GPA 3.93; awarded for 3.90–4.00)
- Coursework: Data Structures, Algorithms, Databases, Operating Systems, Networks, Software Engineering

Notre Dame College Dhaka, Bangladesh
*Higher Secondary Certificate (HSC), GPA **5.00/5.00*** *2021*

Experience

Backend Developer Aug 2025 – Mar 2026
BRACU Cultural Club, BRAC University *Dhaka, Bangladesh*

- Built the club website's entire backend in PHP; awarded a certificate signed by its faculty advisors

Backend Developer Intern (Paid) Jun 2024 – Sep 2025
ATMABISWAS (NGO) *Remote*

- Built the backend of atmabiswas.org single-handedly in PHP and MySQL (PDO): an admin dashboard with session login and CSRF tokens for publishing blogs, notices and job posts with auto-generated job codes
- Hardened file uploads against executable files, protected admin endpoints, sanitized blog HTML with HTML Purifier, and automated deployment to Hostinger with a GitHub Actions workflow over FTPS

Projects

BRACU Routine Builder | *React, Node.js, Express, MongoDB* | GitHub Summer 2026

- Built a full-stack MERN app for planning BRAC University class routines, with paired-day scheduling, two-slot lab spanning and real-time conflict detection that blocks overlapping classes
- Designed a **19-endpoint** REST API in Express and Mongoose with cascading deletes and auto-save, and cached the university's USIS course feed for **7 days** to auto-fill sections, faculty, rooms and exam dates

Process Manager Simulator | *C, POSIX threads, team of 3* | GitHub Spring 2026

- Owned the process lifecycle (fork, exit, wait, kill) and orphan reparenting on a mutex-protected process table, including a blocking wait free of missed-wakeup races

C-Subset Compiler | *C++, Flex, Bison* | GitHub Summer 2026

- Built a compiler front end for a C subset: Flex/Bison parser, scope-aware symbol table of hashed scopes, semantic checks for type, scope and call errors, and a two-pass AST walk that emits three-address code

TrackAlert Vision: Patrol Robot | *C++, Arduino Mega, ESP32-S3-CAM* | GitHub Summer 2026

- **Led a team of 5** as lead architect: designed a two-controller system (Arduino Mega for real-time control, ESP32-S3-CAM for vision and Wi-Fi), and wrote the PD line follower and a Telegram photo-alert bot

Also: Bigger Number Game (Android, Kotlin) · Voting Machine (8086 Assembly) · Arduino Tic-Tac-Toe (C++)

Research

InsightLens: Lecture-to-Notes Pipeline for Code-Mixed Lectures 2026
*Undergraduate thesis, team of 3, grade **A (95.5/100)** | Python, PyTorch, LoRA, Whisper, Qwen2.5-VL*

- Fine-tuned Whisper with LoRA on Bengali-English lectures, cutting character error rate from **67.7% to 15.8%**
- Built the whiteboard-reading stage with Qwen2.5-VL, raising board-content recall from **31.2% to 88.8%**
- S. A. Rafi, A. I. Khan, **A. Habib**, M. G. R. Alam, M. T. Reza, "InsightLens: Generating Checkable Lecture Notes from Code-Mixed Bengali-English Whiteboard Lectures," manuscript in preparation for an IEEE journal

Technical Skills

Languages: Java, JavaScript, PHP, SQL, Python, C, C++, Kotlin, x86 Assembly
Frameworks: Node.js, Express, React, Tailwind CSS, Android (Jetpack Compose)
Databases: MySQL, MongoDB
Developer Tools: Git, GitHub Actions (CI/CD), Vite, Flex/Bison, Arduino, ESP32
Libraries: Mongoose, PDO, POSIX threads, PyTorch